

2023 AMC 10B Problem 5 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 5. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 5

Problem:

Maddy and Lara see a list of numbers written on a blackboard. Maddy adds 3 to each number in the list and finds that the sum of her new numbers is 45. Lara multiplies each number in the list by 3 and finds that the sum of her new numbers is also 45. How many numbers are written on the blackboard?

Answer Choices:

- A. 10
- B. 5
- C. 8
- D. 6
- E. 9

Solution:

Let n be the length of the list and s be the sum of the numbers in the list. Because Lara multiplied each number in the list by 3, the sum of the numbers in the list also increased by a factor of 3. It follows that $s = 15$. Additionally, because Maddy added 3 to each number in the list, the overall sum became $s + 3n = 15 + 3n$. Setting this equal to 45 and solving yields $n = (A) \boxed{10}$. For instance, one possible list is $-3, -2, -1, 0, 1, 2, 3, 4, 5, 6$.

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) ↗

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